Pricing Forecast Capability Assessment of Linear Regression, ARIMA, Random Forest and LSTM Models

Student Name:

Cheng Ying Yu

Ku Ho Kat

Pong Zhun Log

Wong Sai Cheong

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1. Executive Summary

This report evaluates the forecasting capability of four forecasting models, namely Linear Regression, Autoregressive integrated moving average ("ARIMA"), Random Forest, and Long Short-Term Memory ("LSTM"), for price prediction. The models were evaluated using five financial assets, including the S&P 500 Index ("S&P 500"), Hang Seng Index ("HSI"), HSBC Holdings ("0005.HK"), China Petroleum & Chemical Corporation ("0386.HK") and Hong Kong Exchanges and Clearing ("0388.HK") in the report.

The empirical evaluation indicated that forecasting performance varies significantly across forecasting models. Random Forest achieved the best overall performance on three financial assets (i.e., S&P 500, 0388.HK and 0005.HK), while ARIMA delivered the strong results for the S&P 500 Index as well. LSTM outperformed competing models for 0386.HK, while Linear Regression generated the most accurate forecasts for the HSI. These findings suggested that no single model consistently dominated across all market conditions. A hybrid forecasting framework combining ARIMA, Random Forest, and LSTM is therefore recommended for price prediction.

2. Introduction

Price forecasting plays a crucial role in modern portfolio management, risk management, and investment decision-making. Accurate price forecasting enables investors to identify market opportunities, improve asset allocation decisions, and manage downside risks.

The objective of this study is to compares Linear Regression, ARIMA, Random Forest, and LSTM models for forecasting financial asset prices. These models represent statistical, machine learning, and deep learning approaches and are evaluated using five financial assets.

3. Data Analysis & Methodology

3.1 Problem, Assumptions & Limitations

Stock and index price forecasting is extremely difficult due to noise, non-stationarity, random-walk behaviour, regime shifts, and unpredictable events.

Several assumptions underpinned the analysis:

1. Historical market behaviour contains meaningful information that can be utilized for predicting future price movements.
2. Relationships identified in historical data persist over the forecasting horizon.
3. Market prices incorporate information reflected through historical time-series observations.

Despite these assumptions, several limitations exist. Financial markets are influenced by unexpected macroeconomic events, geopolitical developments, regulatory changes, and market sentiment shifts. Consequently, no forecasting model should be regarded as completely accurate .

3.2 Data Collection & Cleansing

Historical data were extracted using the Yahoo Finance (yfinance) API. The dataset consisted approximately three years of closing prices with time-series indexing and excluded future data to avoid look-ahead bias. The 80/20 train-test sequential split of time series data was employed, with earlier 80% of the data for model training and the latter 20% for independent evaluation. Also, data cleansing process focused on ensuring data quality, completeness and consistency.

The below financial assets were selected to represent major indexes and stocks in the market:

- S&P 500 Index (^GSPC)

- Hang Seng Index (^HSI)
- HSBC Holdings Plc (0005.HK)
- Sinopec (0386.HK)
- Hong Kong Exchanges and Clearing (0388.HK)

The data cleaning process focused on ensuring data quality and consistency. Missing observations were handled through backward filling (bfill), followed by removal of any remaining null records (dropna).

3.3 Data Preprocessing

For Linear Regression, extreme data (daily return is greater than 2%~4%) were removed to increase the reliability of estimation.

For ARIMA, stationarity requirements were addressed through differencing transformations. It reduced trend-related biases for autoregressive and moving-average components.

For Random Forest, Bias, Moving Average Divergence, Coefficient of Variation, Trading Activity, and Volatility Momentum were used. These parameters indicated the characteristics of the stock trend, position, and momentum.

For LSTM, Min-Max Scaler was applied to scale all features into the [0, 1] interval, accelerating gradient descent convergence. The baseline LSTM indicated robust capability in modelling volatility clustering and non-linear pricing paths without catastrophic divergence.

Also, data were pre-processed as long-term trends were observed.

- **Moving Averages & Trend Smoothing:** Simple Moving Averages ("SMA") were applied on closing prices to smooth short-term price fluctuations and reveal trend patterns.
- **Correlation Dynamics:** Correlation matrix heatmaps mapping closing prices, daily returns, and moving averages indicate that SMA maintain exceptionally high correlation with the close price (exceeding 90%), while daily returns depict near-zero correlation, aligning with the Efficient Market Hypothesis.
- **Volatility & Stationarity:** Rolling standard deviation analysis confirms time-varying volatility. Augmented Dickey-Fuller ("ADF") stationarity testing on raw series yielded high p-values ($p > 0.05$), confirming non-stationarity and the necessity of prior transformations.

3.4 Model Architecture & Hyperparameter Tuning

The data analysis and models' development were based on Python 3.12.10 with Pandas, NumPy, matplotlib, seaborn, scikit-learn, TensorFlow/Keras, statsmodels, yfinance, etc. For details, please refer to requirements.txt

Four forecasting models were compared for their predictive performance.

- **Linear Regression:** Developed as a benchmark model using historical predictors and lag variables. The model assumed a linear relationship between explanatory variables and future asset prices.
- **ARIMA:** Constructed using optimized autoregressive (p), differencing (d), and moving average (q) parameters. Stationary time-series assumptions were validated prior to model fitting.
- **Random Forest:** Utilized multiple decision trees and ensemble aggregation mechanisms to reduce variance and improve predictive stability. The raw Close series was non-stationary according to the ADF stationarity test, while daily returns were stationary. Consequently, the Random Forest model was trained to forecast

next-day returns rather than absolute price levels, after which predicted returns were converted back to price levels for evaluation. Hyperparameters were selected by GridSearchCV with 5-fold cross-validation and negative mean-squared-error scoring. The parameters n_estimators: {100, 200}, max_depth: {3, 5, 7}, and min_samples_leaf: {20, 30} were employed to search for the best estimator.

- **LSTM:** Employed a deep learning framework utilizing the TensorFlow/Keras Sequential API. The architecture comprises vanilla and/or stacked Long Short-Term Memory layers integrated with Dropout regularization layers to mitigate overfitting, terminating in a fully connected Dense regression layer optimized via Mean Squared Error and the Adam optimizer under chronological train-test partitioning.

Forecasting performance was evaluated using Mean Absolute Error ("MAE"), Root Mean Squared Error ("RMSE") and Mean Absolute Percentage Error ("MAPE"), where lower values indicate higher prediction accuracy.

- **MAE**: Measured the average magnitude of forecasting errors and is relatively robust to outliers.
- **RMSE**: Assigned greater penalties to large prediction errors, making it useful for assessing a model's ability to avoid significant forecasting deviations.
- **MAPE**: Expressed errors as percentages, allowing performance comparisons across assets with different price levels.

4 Results and Discussion

Based on the overall results, Random Forest and ARIMA delivered the most consistent performance for index and stock forecasting, while LSTM demonstrated advantages for selected equities with more complex nonlinear behaviour. Linear Regression provided a useful benchmark but struggled to model nonlinear market dynamics.

Table 1. Overall result

Asset	Best Model	Reason
S&P 500	Random Forest; ARIMA	Lowest MAE and MAPE (Random Forest); Lowest RMSE (ARIMA)
HSI	Linear Regression	Lowest RMSE, MAE and MAPE
0386.HK	LSTM	Lowest RMSE, MAE and MAPE
0388.HK	Random Forest	Lowest RMSE, MAE and MAPE
0005.HK	Random Forest	Lowest RMSE, MAE and MAPE

The detailed performance of each of the five financial assets are tabulated and discussed as below:

Table 2. S&P 500

Model	**RMSE**	**MAE**	**MAPE**
Linear Regression	234.14	193.00	2.68%
LSTM	474.98	468.22	7.65%
Random Forest	223.58	**153.29**	**2.21%**
ARIMA	**222.49**	156.09	2.23%

Note: Best-performing model for each asset is highlighted in blue and bold

ARIMA achieved the lowest RMSE, while Random Forest produced the lowest MAE and MAPE. Overall, both ARIMA and Random Forest demonstrated strong forecasting capability for the S&P 500 Index, whereas LSTM generated substantially larger prediction errors.

Table 3. HSI

Model	RMSE	MAE	MAPE
Linear Regression	**1,438.64**	**1,177.65**	**4.59%**
LSTM	2,351.28	2,311.69	9.67%
Random Forest	2,036.34	1,630.29	6.55%
ARIMA	1,661.94	1,400.72	5.63%

Linear Regression achieved the lowest RMSE, MAE, and MAPE values and therefore provided the best forecasting accuracy among the evaluated models, suggesting that HSI movements during the observation period were relatively well explained by linear trends.

Table 4. 0386.HK

Model	RMSE	MAE	MAPE
Linear Regression	0.45	0.33	6.99%
LSTM	**0.19**	**0.18**	**4.85%**
Random Forest	0.58	0.51	11.77%
ARIMA	0.37	0.32	7.17%

LSTM clearly outperformed the competing models for 0386.HK. It achieved the lowest RMSE, MAE, and MAPE. This result indicated the existence of nonlinear temporal dependencies that are effectively captured by deep learning architectures. Traditional statistical approaches and ensemble models were less successful in modelling the underlying price movements of this stock.

Table 5. 0388.HK

Model	RMSE	MAE	MAPE
Linear Regression	21.59	17.65	4.43%
LSTM	45.53	44.24	11.38%
Random Forest	**14.89**	**12.28**	**3.12%**
ARIMA	30.66	27.26	7.01%

Random Forest achieved the strongest forecasting performance for 0388.HK as it generated the lowest RMSE, MAE, and MAPE , indicating a consistently superior ability to predict stock price movements. The result highlighted the effectiveness of ensemble learning techniques in modelling complex stock market behaviour.

Table 6. 0005.HK

Model	RMSE	MAE	MAPE
Linear Regression	37.83	36.49	25.01%
LSTM	11.25	11.14	12.56%
Random Forest	**8.21**	**6.76**	**4.55%**
ARIMA	13.88	11.82	7.81%

Random Forest delivered the best overall forecasting performance for 0005.HK as it achieved the lowest RMSE, MAE, and MAPE. The consistency of performance across all evaluation metrics suggested that Random Forest effectively captured both linear and nonlinear relationships present in the stock's historical price movements.

5. Recommendations

1. Adopt a model selection approach based on asset type rather than applying a single forecasting methodology across all securities.
2. Use Random Forest as the default production model because it achieved the best overall balance of RMSE, MAE and MAPE.
3. Retain ARIMA as a benchmark and validation model for market indices due to its strong performance on the indexes and stable behaviour.
4. Deploy LSTM selectively for stocks that exhibit nonlinear and temporal dependencies.
5. Incorporate external macroeconomic indicators, sector sentiment indices and volatility metrics to enhance predictive breadth.
6. Adjust training parameters in model tuning and stop with optimal state to prevent overfitting.
7. Further to build a portfolio of weighted assets, monitor and evaluate against market index; track and compare cumulative returns for performance over time.

6. Conclusion

The empirical results demonstrate that forecasting performance is asset-dependent, and that advanced machine learning methods generally outperform traditional linear approaches. Random Forest provided the strongest overall forecasting capability, while ARIMA remained highly competitive for index forecasting. LSTM excelled for selected nonlinear stock series. No single model consistently outperformed all others. Therefore, a hybrid framework combining ARIMA, Random Forest, and LSTM is recommended to improve robustness and forecasting accuracy across different market conditions.

Appendix A. Model Definitions and Theoretical Foundation

Linear Regression

Equation:

$Y = \beta 0 + \beta 1X1 + \beta 2X2 + \ldots + \beta nXn + \varepsilon$

ARIMA

Equation:

$Yt = c + \Sigma \varphi iY(t-i) + \Sigma \theta j\varepsilon(t-j) + \varepsilon t$

Random Forest

Equation (Biau, 2012):

$$\hat{Y} = \frac{1}{B}\sum_{b=1}^{B} T_b(x)$$

B: Total number of trees in the forest.

$T_b(x)$: The prediction output by the b-th individual decision tree.

LSTM

Equation:

$ht = ot \tanh(Ct)$

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